Exercise (Setup)

Make a ~/bin directory (for programs) and add it to your path

% cd // go to your home directory
% mkdir bin // make a subdirectory
% nano .bashrc // edit the startup script
<Add a line “PATH=~/bin:$PATH”>
<start a new terminal>
% echo $PATH // check that ~/bin is in PATH

Download the alignment files we will use:

% cd
% mkdir alignment_files
% cd ~/alignment_files
<Download the files>
% wget http://nucleus.biology.duke.edu/~bredelings/examples.tgz
% tar -zxf examples.tgz // extract the compressed archive
% ls examples // look inside the new directory that was created
% less examples/5S-rRNA/5d.fasta // examine the downloaded files – type q to quit
% alignment-info examples/5S-rRNA/5d.fasta

Move the *fasta-flip.pl* script into ~/bin

% cd ~/alignment_files/examples
% chmod +x fasta-flip.pl // make the file executable
% cp fasta-flip.pl ~/bin // copy the file to ~/bin
% which fasta-flip.pl // check to make sure its in your PATH
% cp 5S-rRNA/25.fasta . // make a copy of a file
% fasta-flip.pl 25.fasta > 25-flipped.fasta // try reversing a FASTA file

Exercise (muscle, HoT)

% cd ~/alignment_files/examples

Make a local copy of a data file:
% cp HIV/HIVSIV.fasta .

Some basic info about the sequences:
% alignment-info HIVSIV.fasta

Generate an alignment using *muscle*:
% muscle < HIVSIV.fasta > HIVSIV-muscle.fasta
% muscle -in HIVSIV.fasta -out HIVSIV-muscle.fasta // This is the same as the previous command.

Reverse the sequences, align them, and then flip them back:
% fasta-flip.pl HIVSIV.fasta > HIVSIV-flipped.fasta // Step 1
% muscle -in HIVSIV-flipped.fasta | fasta-flip.pl > HIVSIV-muscle-flipped.fasta // Steps 2 and 3
% fasta-flip.pl HIVSIV.fasta | muscle | fasta-flip.pl > HIVSIV-muscle-flipped.fasta /// Steps 1, 2, and 3

Look at both alignments:
% seaview HIVSIV-muscle.fasta &
% seaview HIVSIV-muscle-flipped.fasta &

Exercise (Mafft/Probcons/SuiteMSA)

Align sequences and reversed sequences with *mafft*
% mafft HIVSIV.fasta > HIVSIV-mafft-quick.fasta
% mafft --auto HIVSIV.fasta > HIVSIV-mafft.fasta
% fasta-flip.pl HIVSIV.fasta | mafft --auto - | fasta-flip.pl > HIVSIV-mafft-flipped.fasta
% cat HIVSIV.fasta | fasta-flip.pl | mafft --auto - | fasta-flip.pl > HIVSIV-mafft-flipped.fasta

Align sequences and reversed sequences with *probcons*
% probcons HIVSIV.fasta > HIVSIV-probcons.fasta
% fasta-flip.pl HIVSIV.fasta > HIVSIV-flipped.fasta
% probcons HIVSIV-flipped.fasta | fasta-flip.pl > HIVSIV-probcons-flipped.fasta

Compare alignments using SuiteMSA
% SuiteMSA &
<Choose Pixel Plot>
<Click on the left-most "..." to load a new alignment file, and choose HIVSIV-mafft.fasta>
< Also load *HIVSIV-mafft-flipped.fasta*, *HIVSIV-probcons.fasta*, and *HIVSIV-probcons-flipped.fasta* >

Q: What is the spatial pattern of the differences?

Construct an alignment that only contains homologies present in both mafft and probcons alignments
% cat > blank // Type ENTER several times, then press Ctrl-D

% cat blank // It should print some blank lines to the screen.

% cat HIVSIV-mafft.fasta blank HIVSIV-probcons.fasta | alignment-consensus –strict=0.99 > HIVSIV-mafft-probcons.fasta
% seaview HIVSIV-mafft-probcons.fasta &

Exercise (Prank)

Align a (short) 5S data set with *prank*

% cd ~/alignment_files/examples/

% cp 5S-rRNA/25.fasta .

% prank -d=25.fasta -F

% cp output.2.fas prank1.fasta

% prank -d=25.fasta -F

% cp output.2.fas prank2.fasta

% prank -d=25.fasta -F

% cp output.2.fas prank3.fasta

Examine the three files
% seaview prank1.fasta & seaview prank2.fasta & seaview prank3.fasta &

Examine homologies in common between all three prank alignments
% cat prank1.fasta blank prank2.fasta blank prank3.fasta | alignment-consensus –strict=0.99 > prank123.fasta
% seaview prank123.fasta &

Repeat the exercise with HIVSIV.fasta (more interesting, but much slower > 10 minutes)

% prank -d=HIVSIV.fasta -F -o=run1 &

% prank -d=HIVSIV.fasta -F -o=run2 &

<wait for the prank runs to finish>

% cp run1.2.fas HIV-prank1.fasta

% cp run2.2.fas HIV-prank2.fasta

% cat HIV-prank1.fasta blank HIV-prank2.fasta | alignment-consensus –strict=0.99 > HIV-prank12.fasta
% seaview HIV-prank12.fasta &

Exercise

Align sequences with *fsa*. This will take some time.
% cd ~/alignment_files/examples
% fsa --fast --gapfactor 0 HIVSIV.fasta > fsa0.fasta &   // This is similar to probcons – few gaps.
% fsa --fast --gapfactor 0.5 HIVSIV.fasta > fsa0.5.fasta &   // This is the default. Some gaps.
% fsa --fast --gapfactor 1 HIVSIV.fasta > fsa1.fasta &   // This is extra conservative, more gaps.

Look at the alignment lengths for the different *fsa* parameters:
% alignment-info fsa0.fasta
% alignment-info fsa0.5.fasta
% alignment-info fsa1.fasta

Q: How do the lengths of the 3 fsa alignments differ?

Q: How do the lengths compare to the alignment lengths for other programs?